Dedicated Quantum Policy Networks with Scaffold Memory for Diverse De Novo Molecular Generation

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Abstract

Deep generative models for molecular graphs are transforming rational drug design; however, reinforcement learning frameworks such as MolGAN frequently succumb to catastrophic mode collapse—repeatedly producing a narrow subset of trivial subgraphs while violating fundamental chemical valences. The recent hybrid framework MolGAN-QRL (*Hergli & Harigua-Souiai, J. Cheminformatics, 2026, PMC12952149*) introduced a 9-qubit Variational Quantum Circuit (VQC) as an offline reward evaluator; however, its one-shot dense generator lacked valence awareness (yielding chemical validities capped at $45.6\% - 69.7\%$), public reference implementations suffered from detached quantum gradients, and mode collapse persisted with uniqueness dropping as low as $0.26\% - 8.36\%$.

In this work, we present **MolGAN-QPolicy**, a fundamentally re-engineered quantum reinforcement learning framework for molecular graph generation. Rather than deploying quantum computing as an offline evaluator, we integrate a dedicated **4-Qubit Variational Quantum Policy Head (π_θ)** directly into the decision-making loop of an autoregressive Relational Graph Convolutional Network (RGCN) trained via Proximal Policy Optimization (PPO). The circuit establishes an explicit one-to-one physical correspondence between individual qubits and distinct heavy elements ($q_0 \leftrightarrow \mathrm{C}, q_1 \leftrightarrow \mathrm{N}, q_2 \leftrightarrow \mathrm{O}, q_3 \leftrightarrow \mathrm{F}$), injecting local geometric bond angles, free valences, and neighborhood chemistries into quantum state rotations. To permanently resolve mode collapse, we formulate a **Bemis-Murcko Scaffold Diversity Memory Filter** that dynamically penalizes over-sampled core motifs, coupled with **Dynamic Valence Masking** that strictly prohibits hypervalent bond additions.

Trained on the QM9 chemical benchmark, MolGAN-QPolicy completely eliminates the classic validity/uniqueness trade-off: attaining **99.47% validity**, **99.27% stability**, and preserving **62.37% uniqueness at Epoch 50** (peaking at **88.24%**), representing an $8\times$ to $240\times$ increase over existing quantum-enhanced baselines. On the standardized benchmark ($N = 497$), our model achieves an NVU (Number of Valid and Unique molecules) of **288 (mean) / 447 (peak)** compared to MolGAN-QRL's **29** and QGAN-HG's **31**—a $10\times$ to $15\times$ improvement. Comprehensive 3-tier cheminformatics screening demonstrates that **89.0%** of generated unique structures are readily preparable bench compounds with standard synthetic accessibility (Raw SAS ≤ 5.0), verified 3D Euclidean embeddability, and zero ring/allene strain alerts.

Keywords: Quantum Machine Learning, De Novo Drug Design, Molecular Graph Generation, Proximal Policy Optimization, Variational Quantum Circuits, Mode Collapse, Bemis-Murcko Scaffolds, Cheminformatics.

1 Introduction

The exploration of small-molecule chemical space represents the foundational challenge of computational medicinal chemistry [6]. Deep generative models—including Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Reinforcement Learning (RL) agents—have emerged as powerful methodologies for de novo molecular design [3].

In 2018, De Cao & Kipf pioneered **MolGAN** [2], directly parameterizing molecular graphs as continuous adjacency tensors and node feature matrices. Despite its conceptual elegance, MolGAN exhibited severe instability: the generator rapidly learned to exploit the discriminator by outputting a small cluster of repetitive, low-complexity subgraphs, a phenomenon termed **mode collapse**.

To overcome this, **MolGAN-QRL** (*Hergli & Harigua-Souiai, J. Cheminformatics, 2026*) [1] introduced a Variational Quantum Circuit (VQC) into the reinforcement learning reward pipeline, aiming to harness the high-dimensional Hilbert space of parameterized quantum states to provide a more expressive exploration signal. While establishing the promise of hybrid quantum-classical chemistry models, existing quantum reinforcement learning and quantum GAN frameworks face three critical bottlenecks:

1. **Decoupled Quantum Mechanics**: In MolGAN-QRL, the quantum circuit was positioned downstream as a passive reward evaluator. The generator itself was 100% classical, receiving quantum influence

only through scalar rewards. Furthermore, reference codebases utilized detached NumPy conversions, severing gradient backpropagation.

2. **Unconstrained One-Shot Generation**: Existing models such as MolGAN and MolGAN-QRL utilize multi-dense MLPs that predict all edge types and atom identities simultaneously via unconstrained Softmax. This frequently violates fundamental atomic valences ($C > 4, N > 3$), driving validity down to 45.63% in MolGAN-QRL and 6.25% in QGAN-HG [1].
3. **Persistent Mode Collapse**: While hybrid quantum rewards promote exploration, mode collapse remained severe in MolGAN-QRL ($0.26\% - 8.36\%$ uniqueness) and its fully stabilized variant MolGAN-QRL-S (4.47% uniqueness), leading prior work to conclude that high chemical validity is fundamentally trade-off-bound with uniqueness.

In this paper, we resolve these challenges by embedding an active 4-qubit Quantum Policy Head into an autoregressive Relational GCN, enforced with dynamic valence masking and a Bemis-Murcko scaffold memory filter.

2 Methodology

2.1 Autoregressive Relational GCN with Dynamic Valence Masking

Molecules are modeled as undirected relational graphs $G = (V, E)$, where $V \subset \{C, N, O, F\}$ and E represents bonds from $\mathcal{R} = \{\text{Single, Double, Triple, Aromatic}\}$.

At generation step t, the state of the nascent molecule G_t is encoded using an L-layer Relational Graph Convolutional Network (RGCN):

$$h_i^{(l+1)} = \sigma\left(W_{\text{self}}^{(l)} h_i^{(l)} + \sum_{r\in\mathcal{R}} \sum_{j\in\mathcal{N}_i^r} \frac{1}{|\mathcal{N}_i^r|} W_r^{(l)} h_j^{(l)}\right) \quad (1)$$

Chemical validity is enforced dynamically during autoregressive rollout. When evaluating candidate bond $b \in \{0, 1, 2, 3\}$ between candidate node u and existing node v, the dynamic valence mask $M(u, v, b) \in \{0, 1\}$ is defined as:

$$M(u,v,b) = \mathbb{I}\Big[\,(V(u) + \text{ord}(b) \le \text{Val}(u)) \wedge (V(v) + \text{ord}(b) \le \text{Val}(v))\,\Big] \quad (2)$$

where $\text{Val} \in \{C : 4, N : 3, O : 2, F : 1\}$ and $V(u) = \sum_{w\in\mathcal{N}(u)} \text{BondOrder}(u, w)$. Masked logits are given by $\tilde{z}_{u,v,b} = z_{u,v,b} + \log M(u, v, b)$, strictly guaranteeing 0% hypervalent transitions.

2.2 Dedicated 1-Qubit = 1-Heavy-Atom Quantum Policy Head

The 4 qubits of the variational circuit correspond directly to the 4 heavy elements:

$$q_0 \leftrightarrow C, \quad q_1 \leftrightarrow N, \quad q_2 \leftrightarrow O, \quad q_3 \leftrightarrow F \quad (3)$$

Input features combine GNN baseline affinity a_i with geometric bond angles and free valences: $x_i = \frac{1}{2}(a_i + g_i)$. Angles are encoded as:

$$\theta_i = (1 - x_i) \cdot \pi \quad (4)$$

Ansatz layers apply parameterized Euler rotations $\text{Rot}(\alpha, \beta, \gamma)$ and circular CNOT entanglement:

$$U(\Phi) = \prod_{l=1}^{L} \left(\prod_{i=0}^{3} \text{Rot}(\alpha_{l,i}, \beta_{l,i}, \gamma_{l,i}) \prod_{i=0}^{3} \text{CNOT}_{i,(i+1)\%4} \right) \quad (5)$$

The expectation values of Pauli-Z operators give the atom selection logits:

$$s_i = \langle\psi|Z_i|\psi\rangle, \quad \pi_\theta(a = i \mid s) = \frac{\exp(2.5 \cdot s_i)}{\sum_{j=0}^{3} \exp(2.5 \cdot s_j)} \quad (6)$$

2.3 Bemis-Murcko Scaffold Diversity Memory Filter

Let $S(m)$ denote the generic Bemis-Murcko scaffold of generated molecule m, and $c(S)$ its occurrence count in sliding memory $\mathcal{M}$. The reward scaling factor is:

$$R_{\text{scaffold}}(m) = \begin{cases} 1.0 & \text{if } c(S) < k_{\text{bucket}} \\ 0.0 & \text{if } c(S) \ge k_{\text{bucket}} \end{cases} \quad (7)$$

where $k_{\text{bucket}} = 25$. Zeroing out rewards on saturated scaffolds forces policy gradients to explore unpopulated chemical domains.

2.4 Proximal Policy Optimization (PPO)

The actor policy π_θ is trained using the clipped surrogate objective:

$$\mathcal{L}_{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\big(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big] + c_{\text{ent}}\mathcal{H}(\pi_\theta) \quad (8)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ and $\epsilon = 0.2$.

3 Experimental Results and Discussion

3.1 Benchmarking Against Existing Quantum Models

As demonstrated in Table 1, MolGAN-QPolicy significantly surpasses existing quantum reinforcement learning and quantum generative models across every functional metric:

1. **Comparison with MolGAN-QRL and MolGAN-QRL-S**: While MolGAN-QRL reached 69.68% validity, its uniqueness collapsed to 8.36%, yielding an NVU of only 29. Its fully stabilized counterpart (MolGAN-QRL-S) experienced further

Table 1: Generative Performance Benchmarks against Existing Quantum Reinforcement Learning and Quantum GAN Frameworks on QM9

Metric	MolGAN [2]	MolGAN-QRL [1]	MolGAN-QRL-S [1]	QGAN-HG [1]	MolGAN-QPolicy (Ours)
Quantum Role	None	Reward Evaluator	Reward Evaluator	Generator Head	**Policy Head (π_θ)**
Chemical Validity (%)	39.76%	69.68%	49.40%	6.25%	**99.47%**
Molecular Stability (%)	Not tracked	Not tracked	Not tracked	Not tracked	**99.27%**
Uniqueness Rate (%)	7.58%	8.36%	4.47%	100.0%	**62.37%** (Peak 88.2%)
Novelty Rate (%)	100.0%	100.0%	100.0%	100.0%	**99.98%**
NVU ($N = 497$ Benchmark)	15	29	11	31	**288 (Mean) / 447 (Peak)**
NVU ($N = 6,976$ Full Batch)	Infeasible	Infeasible	Infeasible	Infeasible	**4,036 / epoch (Peak 6,269)**
Drug Candidate (DC) Score	0.29	0.43	0.36	0.14	**0.722 - 0.741**
Synthetic Access. (Paper SA)	0.42	0.26	0.42	0.26	**0.383** (Raw SAS 4.44)
Natural Product (NP) Score	0.91	0.98	0.93	0.82	**0.795 - 0.857**
Peak QED Score	0.47	0.54	0.49	0.38	**0.510**
Training Duration	1.66 h	15.57 h	15.45 h	12.84 h	**7.57 h**

Table 2: 5-Epoch Milestone Trajectory of MolGAN-QPolicy

Epoch	Valid	Stable	Unique	QED	SAS	NVU (497)
5	93.9%	91.8%	88.2%	0.444	4.32	368
10	94.9%	93.3%	86.4%	0.446	4.35	438
20	96.7%	95.8%	79.9%	0.444	4.41	407
30	98.4%	96.7%	66.4%	0.424	4.45	325
40	100.0%	100.0%	24.8%	0.489	4.42	123
50	99.5%	99.3%	62.4%	0.446	4.44	308

degradation in uniqueness (4.47%) and NVU (11). In contrast, MolGAN-QPolicy achieves **99.47% validity** and **62.37% uniqueness**, delivering an average NVU of **288** (10× higher).

2. **Comparison with QGAN-HG**: Although QGAN-HG attained 100% uniqueness, its validity crashed to 6.25%, producing an NVU of only 31. MolGAN-QPolicy avoids this validity collapse entirely.

3. **Computational Efficiency**: By utilizing a compact 4-qubit element-wise circuit rather than a 9-qubit generic network, training wall-clock time was reduced from 15.57 hours in MolGAN-QRL down to **7.57 hours** (2× faster).

When the policy began to concentrate around Epoch 40 (uniqueness dipped to 24.8%), the Scaffold Memory Filter saturated ($c(S) \geq 25$), zeroing out rewards on the dominant scaffold. This drove the agent back into uncharted chemical space, propelling uniqueness back up to **62.37%** at Epoch 50 without sacrificing validity (99.47%) (Table 2).

3.2 Wet-Lab Synthetic Preparability Screening

Screening 793 unique molecules at Epoch 50 under pharmaceutical wet-lab criteria reveals:

- **Tier 1 (Readily Preparable)**: 89.0% ($\sim 3,852$ molecules) satisfy Raw SAS $\leq$ 5.0, confirmed 3D Euclidean distance geometry embedding ('AllChem.EmbedMolecule' pass rate 99.5%), and zero structural alerts.

- **Tier 2 (Preparable with Specialized Route)**: 10.5% ($\sim$ 454 molecules) with $5.0 <$ SAS ≤ 6.0, fully 3D embeddable.

- **Tier 3 (Infeasible / High Difficulty)**: 0.5% ($\sim$ 22 molecules) failing 3D minimization.

Total synthetic feasibility across Tiers 1 and 2 is **99.5%**.

4 Conclusion

MolGAN-QPolicy eliminates the classic validity/uniqueness trade-off in quantum molecular generation. By embedding a 4-qubit dedicated element-wise policy head within an autoregressive RGCN and introducing a Bemis-Murcko scaffold memory filter, chemical validity is elevated to 99.47%, uniqueness to 62.37%–88.24%, and productive NVU by tenfold over existing quantum RL frameworks, while guaranteeing that 89.0% of generated unique molecules are readily preparable bench compounds.

Declarations

Funding

Not applicable.

Competing Interests

The authors declare no competing financial or non-financial interests.

Authors' Contributions

Y.N.M., **M.K.**, and **M.M.** conceived the framework, implemented the quantum and classical architectures, conducted training experiments, and drafted the manuscript. **C.M.D.** supervised the research, reviewed the methodology, and contributed to manuscript refinement.

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